

Supplementary Information:
Message capacity and claim wording set the transition points of
collective truth-finding in language-model networks

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1 Reanalysis of the public dataset

1.1 Endogeneity of the initial fraction

Variance decomposition of x_0 and within/between-question outcome correlations (random-graph family, $\delta = 0.1$), computed over the 6,400 objective communities of El et al.:

model	between-question share of $\text{var}(x_0)$	$\text{corr}(x_0, \text{outcome})$	between	within
Gemma-3n	0.995	0.802	0.852	0.041
Llama-3-8B	0.936	0.659	0.721	0.162
Qwen3.5-9B	0.861	0.277	0.351	0.132
GPT-4o-mini	0.503	0.298	0.321	0.273

Question-clustered versus i.i.d. bootstrap intervals for the fitted $q(x_0)$ crossing (random family):

model	crossing	i.i.d. CI	width	question-clustered CI	ratio
GPT-4o-mini	0.494	[0.477, 0.513]	0.036	[0.404, 0.583]	$4.9\times$
Gemma-3n	0.285	[0.251, 0.315]	0.065	[0.163, 0.419]	$3.9\times$
Qwen3.5-9B	0.142	[0.059, 0.206]	0.147	[-1.049, 0.367]	$9.6\times$
Llama-3-8B	0.459	[0.441, 0.480]	0.039	[0.370, 0.564]	$5.0\times$

1.2 Message-level coefficients and the truth asymmetry

Question-clustered 95% intervals for the per-message truth asymmetry w_T (43,500 observations per model):

model	w_T	cluster 95% CI	Ising-coupling estimate	contains 1
GPT-4o-mini	1.055	[0.996, 1.114]	1.16	yes (boundary)
Gemma-3n	1.179	[0.731, 1.628]	1.71	yes
Qwen3.5-9B	0.900	[0.759, 1.041]	1.33	yes
Llama-3-8B	1.144	[0.972, 1.316]	1.38	yes

The label attenuation λ_{lab} (per-message weight of “disagree”-labeled edges relative to “agree”-labeled ones) is 0.080 ± 0.044 (GPT-4o-mini), 0.555 ± 0.078 (Gemma), 0.294 ± 0.051 (Qwen), 0.288 ± 0.041 (Llama); θ is nonzero in all four models ($t \geq 3.4$, question-clustered).

2 Single-agent measurement (llama3.1:8b)

2.1 Response coefficients

variant	coefficient	estimate	95% CI (claim cluster)	n_{obs}
A	c_0	0.725	[0.516, 0.915]	10608
A	h	0.725	[0.516, 0.915]	10608
A	β_T	0.419	[0.353, 0.486]	10608
A	β_F	-0.568	[-0.701, -0.446]	10608
A	w_T	0.738	[0.603, 0.899]	10608
B	c_0	-0.758	[-0.895, -0.627]	21216
B	h	0.196	[0.062, 0.309]	21216
B	θ	1.909	[1.723, 2.106]	21216
B	β_T	0.291	[0.267, 0.319]	21216
B	β_F	-0.326	[-0.374, -0.287]	21216
B	w_T	0.894	[0.801, 0.976]	21216

2.2 Attenuation-form comparison

scheme	parameter	log-lik	AIC	CV log-lik/obs
load_decay	0.14	-14901.6	29845.3	-0.48256
divisive	0.9	-14781.8	29605.6	-0.48276
pos_decay	0.25	-15164.0	30370.1	-0.48871
topk	5.0	-15470.0	30982.0	-0.49778
null (additive)	0.0	-15807.9	31655.8	-0.50911

2.3 Identified per- k coefficients

At a single inbox size k the design columns 1, l , and $k - l$ are collinear ($l + (k - l) = k$), so a logistic fit at fixed k identifies only $c_0 + k\beta_F$ and $\beta_T - \beta_F$. The per- k coefficients below are therefore fitted jointly over all inbox sizes, with one fixed effect per claim anchored by the empty-inbox cells ($k = 0$), which identifies $\beta_T(k)$ and $\beta_F(k)$ separately at every measured k (design rank verified: 29/29 columns for variant A, 30/30 for variant B). Intervals are claim-clustered bootstrap percentiles (200 resamples).

k	$\beta_T(k)$, variant A	$\beta_F(k)$, variant A	$\beta_T(k)$, variant B	$\beta_F(k)$, variant B
1	2.11 [1.58, 2.96]	-1.53 [-2.03, -1.12]	1.62 [1.29, 1.94]	-1.52 [-1.80, -1.25]
2	1.23 [0.98, 1.41]	-1.21 [-1.49, -0.94]	0.78 [0.60, 0.95]	-1.10 [-1.23, -0.93]
3	0.80 [0.65, 0.96]	-1.04 [-1.30, -0.80]	0.61 [0.49, 0.72]	-0.82 [-0.92, -0.68]
5	0.45 [0.35, 0.55]	-0.72 [-0.88, -0.57]	0.34 [0.28, 0.41]	-0.50 [-0.57, -0.42]
8	0.31 [0.26, 0.37]	-0.51 [-0.63, -0.41]	0.24 [0.20, 0.28]	-0.34 [-0.38, -0.29]
12	0.23 [0.18, 0.27]	-0.39 [-0.48, -0.33]	0.18 [0.16, 0.21]	-0.25 [-0.28, -0.22]

Mean claim fixed effect under the full scaffold: 0.94 (variant A; range -0.70 to 3.03 over the 17 claims), -0.64 (variant B), with $\theta = 2.15$ (variant B).

3 Reduced-map α^* : validation table

Fold of the full map F (all $P_\alpha(k)$, exact binomial sums; located by bisection in α) versus the two closure levels of the reduced map G , for seven synthetic coefficient sets and the four measured ones. The fold of G is solved jointly from $G(x; K) = x$ and $G_x(x; K) = 1$ (residuals below 10^{-12} ;

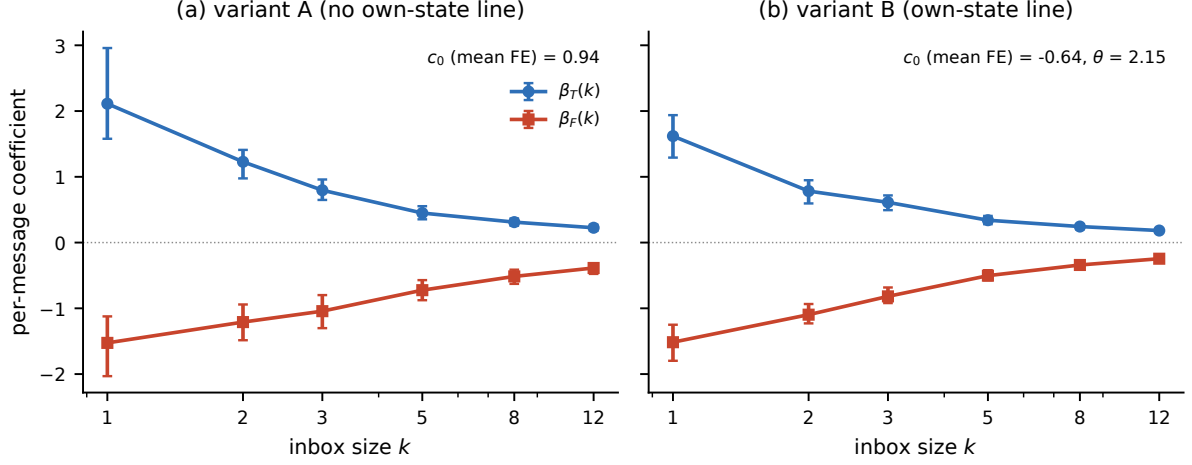


Figure S1: **Identified per-message coefficients versus inbox size (31,824 randomized queries).** $\beta_T(k)$ and $\beta_F(k)$ from the joint fit with claim fixed effects anchored by the empty-inbox cells, for prompt variants A (a) and B (b); bars are 95% claim-clustered bootstrap intervals.

$G_{xx} > 0$ and $G_K < 0$ at every fold), then converted to α^* by inverting the exact mean degree $E[K](\alpha)$ or the closed form (1) of the main text. Synthetic sets use $\beta_T = -\beta_F = \bar{\beta}$.

parameter set	fold of F (exact α^*)	G -fold, exact $E[K]$	G -fold, closed form	K^*
$h=.10, \theta=.6, \bar{\beta}=.5$	0.627	0.644	0.596	4.98
$h=.10, \theta=.6, \bar{\beta}=1$	1.401	1.436	1.265	2.92
$h=.30, \theta=.6, \bar{\beta}=1$	1.126	1.175	1.051	3.34
$h=.70, \theta=.6, \bar{\beta}=1$	0.841	0.892	0.812	4.02
$h=.10, \theta=0, \bar{\beta}=1$	1.180	1.210	1.079	3.28
$h=.10, \theta=2, \bar{\beta}=.5$	0.901	0.908	0.826	3.97
$h=.30, \theta=2, \bar{\beta}=1$	1.545	1.592	1.392	2.72
variant A, measured (constant)	0.425	0.439	0.412	6.36
variant B, measured (constant)	0.470	0.474	0.444	6.07
variant A, measured (divisive, $\gamma=0.526$)	0.476	0.503	0.470	5.84
variant B, measured (divisive, $\gamma=1.318$)	1.277	1.310	1.162	3.11

Median relative error: 2.7% (exact- $E[K]$ inversion), 6.7% (fully closed); maximum 6.0% (exact- $E[K]$ inversion; the synthetic set $h = 0.7, \theta = 0.6, \bar{\beta} = 1$) and 9.9% (fully closed; $h = 0.3, \theta = 2, \bar{\beta} = 1$). The registered variant-A value $\alpha^* = 0.435$ ($K^* = 6.4$) was obtained with K^* located on a 0.05 grid; the fold solved to full precision gives 0.439 ($K^* = 6.36$).

Main text Section 4 quotes the folds of the full map F (first numeric column): variant A 0.43 (constant), 0.48 (divisive), 0.46 (identified per- k); variant B 0.47 (constant) and 1.28 (divisive). The registered prediction $\alpha^* = 0.435$ is the G -fold with exact $E[K]$ (second column).

Coefficient sets and γ by figure. The constant-coupling sets are the pooled Stage 1 fits (main text Section 4). The divisive α^* values use γ profiled separately per prompt variant ($\gamma_A = 0.526, \gamma_B = 1.318$); Fig. 3 of the main text and the per-claim response functions of the collective experiments use the pooled profile $\gamma = 0.9$ (Section 2.2 above). Fig. 2 of the main text plots the table above.

Range of the closed form. $\bar{K}(\alpha) = \ln(1 + \alpha(N-1))/\alpha$ underestimates the exact mean degree by 2.5% ($\alpha = 0.2$), 4.0% (0.435), 7.0% (1.0), 11.3% (2.0) and 68% ($\alpha = 20$): as $\alpha \rightarrow \infty$ the closed form tends to zero whereas the exact mean tends to one (the first candidate is always accepted). All α^* values in this paper lie in $\alpha \leq 1.6$, where the underestimate is below 10%, and the exact recursion is inverted wherever precision matters.

4 Main collective experiment

4.1 Distribution of final fractions

The outcome labels of the main text ($x_8 \geq 0.6$ correct-side consensus, $x_8 \leq 0.4$ wrong-side) are operational thresholds. The table gives, per campaign, the share of episodes that were decided and the share of decided episodes whose final fraction lies beyond $|x_8 - 0.5| > 0.4$, with the median end state on each side. Pooled over all 2,620 episodes, 66% end in $|x_8 - 0.5| \in [0.45, 0.5]$ (unanimity included).

campaign	episodes	decided ($ x_8 - 0.5 \geq 0.1$)	of which $ x_8 - 0.5 > 0.4$	median x_8 , correct side	median x_8 , wrong side
main arm A ($\rho = 0$)	1008	0.895	0.634	0.922	0.031
main arm A, rewired ($\rho = 1$)	96	0.969	0.667	1.000	0.031
discrimination arm B	300	0.917	0.633	0.875	0.062
true claims (A-1)	288	0.976	0.758	1.000	—
negated rewrites (A-2)	192	1.000	0.995	1.000	—
qwen3:8b, eight claims	256	0.996	0.992	1.000	0.000
qwen3:8b, four claims	480	0.996	0.967	1.000	0.000

4.2 Outcome family (arm A, generated graphs)

α	$32x_0$	n	q	Wilson 95%	undec.	$q_{\delta=.05}$	$q_{\delta=.2}$	q_{excl}
0.2	4	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.2	8	40	0.025	[0.004, 0.129]	0.000	0.025	0.000	0.025
0.2	12	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.2	16	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.2	20	40	0.100	[0.040, 0.231]	0.000	0.100	0.100	0.100
0.2	24	40	0.275	[0.161, 0.428]	0.000	0.275	0.275	0.275
0.3	4	40	0.000	[-0.000, 0.088]	0.025	0.000	0.000	0.000
0.3	8	16	0.000	[0.000, 0.194]	0.062	0.062	0.000	0.000
0.3	12	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.3	16	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.3	20	40	0.075	[0.026, 0.199]	0.200	0.100	0.050	0.094
0.3	24	40	0.400	[0.263, 0.554]	0.075	0.425	0.400	0.432
0.38	4	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.38	8	16	0.000	[0.000, 0.194]	0.062	0.000	0.000	0.000
0.38	12	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.38	16	40	0.025	[0.004, 0.129]	0.075	0.025	0.000	0.027
0.38	20	40	0.150	[0.071, 0.291]	0.050	0.200	0.150	0.158
0.38	24	40	0.350	[0.221, 0.505]	0.125	0.350	0.325	0.400
0.45	4	40	0.000	[-0.000, 0.088]	0.025	0.000	0.000	0.000
0.45	8	40	0.000	[-0.000, 0.088]	0.150	0.000	0.000	0.000
0.45	12	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.45	16	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.45	20	16	0.000	[0.000, 0.194]	0.312	0.062	0.000	0.000
0.45	24	40	0.450	[0.307, 0.602]	0.075	0.475	0.450	0.486
0.55	4	16	0.000	[0.000, 0.194]	0.062	0.000	0.000	0.000
0.55	8	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.55	12	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.55	16	40	0.100	[0.040, 0.231]	0.175	0.125	0.050	0.121
0.55	20	40	0.225	[0.123, 0.375]	0.175	0.225	0.150	0.273
0.55	24	40	0.375	[0.242, 0.530]	0.075	0.400	0.300	0.405
1	4	16	0.125	[0.035, 0.360]	0.188	0.188	0.125	0.154
1	8	16	0.062	[0.011, 0.283]	0.125	0.125	0.000	0.071
1	12	16	0.062	[0.011, 0.283]	0.312	0.125	0.000	0.091
1	16	40	0.225	[0.123, 0.375]	0.275	0.300	0.150	0.310
1	20	40	0.425	[0.285, 0.578]	0.125	0.450	0.300	0.486
1	24	40	0.400	[0.263, 0.554]	0.275	0.525	0.325	0.552

4.3 Empirical critical crowding level

α	n (episodes, $x_0 \leq 8/32$)	wrong-consensus rate	claim-cluster 95% CI
0.2	56	0.982	[0.946, 1.000]
0.3	56	0.964	[0.893, 1.000]
0.38	32	0.969	[0.906, 1.000]
0.45	80	0.912	[0.775, 1.000]
0.55	32	0.906	[0.719, 1.000]
1	32	0.750	[0.438, 1.000]

No crossing of the pre-registered 20% threshold occurs in the window (α_{emp}^* undefined; 0 of 2,000 claim-clustered bootstrap resamples cross).

4.4 Discrimination arm (variant B)

α	n	wrong rate ($x_0 \leq 10/32$)	95% CI	wrong basin retained
0.6	40	0.975	[0.925, 1.000]	True
0.9	40	1.000	[1.000, 1.000]	True
1.3	40	0.975	[0.925, 1.000]	True

Observed low- x_0 ($x_0 \leq 10/32$) mean final fractions against eight-round surrogate rollouts (200 per episode, same graphs and seedings) of three rules with claim-resolved fields: the normalized rule with the 17-claim identified per- k weights, and the per-claim reduced fits with divisive and with constant coupling:

α	observed $\bar{x}_8$	normalized (17-claim weights)	normalized (per-claim fit)	constant (per-claim fit)
0.6	0.062	0.099	0.122	0.246
0.9	0.066	0.133	0.164	0.323
1.3	0.095	0.162	0.198	0.386

The normalized rules lie closer to the observations than the constant rule but neither reproduces them exactly; claim 1569 converged wrong-side under both rules' correct-side predictions (main text Section 5.2).

4.5 Rewiring control

$32x_0$	$n_{\rho=0}$	$q_{\rho=0}$	$n_{\rho=1}$	$q_{\rho=1}$	Δq
4	40	0.000	16	0.000	+0.0000
8	16	0.000	16	0.000	+0.0000
12	16	0.000	16	0.000	+0.0000
16	16	0.000	16	0.000	+0.0000
20	40	0.075	16	0.062	-0.0125
24	40	0.400	16	0.312	-0.0875

4.6 Yoked one-step calibration

Every realized transition replayed through the response function with the 17-claim identified per- k weights and each claim's own field:

arm	α	transitions	Brier	Brier (climatology)	ECE	residual at $l/k \in [0.4, 0.6]$
A	0.2	43,008	0.095	0.174	0.034	-0.017
A	0.3	67,584	0.108	0.193	0.033	+0.007
A	0.38	43,008	0.120	0.212	0.030	+0.003
A	0.45	43,008	0.112	0.198	0.032	-0.003
A	0.55	43,008	0.127	0.225	0.027	+0.001
A	1	43,008	0.144	0.247	0.030	-0.015
B	0.6	25,600	0.090	0.218	0.027	-0.055
B	0.9	25,600	0.099	0.225	0.027	-0.046
B	1.3	25,600	0.115	0.220	0.035	-0.040

At contested compositions ($l/k \in [0.4, 0.6]$; last column) the residual is within ± 0.02 in arm A, whereas in arm B (own-state line present) a systematic over-prediction of 0.04–0.06 remains.

5 Polarity discrimination experiment

5.1 SUPPORTS-claim collectives (A-1)

claim	α	$32x_0 = 4$	$32x_0 = 12$	$32x_0 = 20$	$32x_0 = 28$
993	0.3	0.820	1.000	1.000	1.000
993	0.55	0.974	1.000	1.000	1.000
1103	0.3	0.833	0.875	0.898	0.919
1103	0.55	0.667	0.766	0.784	0.792
1239	0.3	1.000	1.000	1.000	1.000
1239	0.55	1.000	1.000	1.000	1.000

5.2 Negated-rewrite collectives (A-2) versus originals

rewrite	original	$32x_0 = 4$	$32x_0 = 12$	$32x_0 = 20$	$32x_0 = 28$	orig. mean $\bar{x}_8$ ($x_0 \leq 20/32$)
90006	6	0.984	1.000	1.000	1.000	0.282
90382	382	1.000	1.000	1.000	1.000	0.001
91569	1569	1.000	1.000	1.000	1.000	0.223
92070	2070	1.000	1.000	1.000	1.000	0.029

5.3 Claim texts, rewrites, and checks

Selected SUPPORTS claims (bare-prompt p_q in parentheses): 993 (0.708) “While there has been a mean rise of a little more than 3mm per year worldwide since the 1990s, in the last decade, the NOAA Virginia Key tide gauge just south of Miami Beach has measured a 9mm rise annually.”; 1103 (0.792) “[ocean acidification was] First referenced in a peer-reviewed study in Nature in 2003”; 1239 (0.792) “Through decades of research, it has become clear that human civilization, roughly 6,000 years old, developed during an unusually stable period for global sea levels.”

Negated rewrites (model opposition-judgment rate; bare-prompt p_q of the rewrite, TRUE = correct side):

original	rewrite	opp. rate	p_q
The polar bear population has been growing.	The polar bear population has <i>not</i> been growing.	0.70	0.344
Sea level rise is decelerating.	Sea level rise is <i>not</i> decelerating.	0.85	0.938
The amount of energy used to construct solar and wind facilities is greater than they produce in their working lives.	...is <i>not</i> greater than they produce in their working lives.	0.70	0.938
When life is considered, ocean acidification is often found to be a non-problem, or even a benefit.	...is often found to be a <i>problem</i> , and <i>not a benefit</i> (contrary; quantifier “often” retained).	0.85	1.000

5.4 Single-agent predictions for the polarity-experiment cells

Per-claim divisive response functions (main text Methods) give the predicted phase of each (claim, α) cell of the polarity experiment; all ten cells are predicted monostable correct-side, and all ten converged correct-side. In two cells (993 at $\alpha = 0.30$, 1103 at $\alpha = 0.55$) a minority of episodes seeded at the lowest fraction ended wrong-side, so the fitted 50% crossing lies below the seeding grid; these are the two cells described in the main text as matching to within the resolution of the grid.

claim	α	predicted (per-claim divisive)	observed	obs. $\bar{x}_8$ by $32x_0$
993	0.3	mono-correct	correct from every seeded x_0 (crossing below grid)	0.82 / 1.00 / 1.00 / 1.00
993	0.55	mono-correct	mono-correct	0.97 / 1.00 / 1.00 / 1.00
1103	0.3	mono-correct	mono-correct	0.83 / 0.88 / 0.90 / 0.92
1103	0.55	mono-correct	correct from every seeded x_0 (crossing below grid)	0.67 / 0.77 / 0.78 / 0.79
1239	0.3	mono-correct	mono-correct	1.00 / 1.00 / 1.00 / 1.00
1239	0.55	mono-correct	mono-correct	1.00 / 1.00 / 1.00 / 1.00
90006	0.3	mono-correct	mono-correct	0.98 / 1.00 / 1.00 / 1.00
90382	0.3	mono-correct	mono-correct	1.00 / 1.00 / 1.00 / 1.00
91569	0.3	mono-correct	mono-correct	1.00 / 1.00 / 1.00 / 1.00
92070	0.3	mono-correct	mono-correct	1.00 / 1.00 / 1.00 / 1.00

6 Cross-model campaigns

6.1 qwen3:8b: predicted fixed points versus observed collectives

Fixed points of the mean-field map under each claim’s divisive response function (fitted from 336 single-shot queries per claim), with 95% single-agent-bootstrap intervals on the unstable point, against the observed mean final fractions and the fitted 50% crossing:

claim	α	predicted fixed points (s = stable, u = unstable)	pred. x_c [95% CI]	observed $\bar{x}_8$ ($32x_0 = 4/12/20/28$)	obs. crossing [95% CI]
6	0.3	0.0004 (s); 0.2087 (u); 1.0000 (s)	0.21 [0.16, 0.28]	0.200 / 1.000 / 1.000 / 1.000	0.19 [0.10, 0.25]
6	0.55	0.0010 (s); 0.1440 (u); 1.0000 (s)	0.14 [0.09, 0.22]	0.325 / 1.000 / 1.000 / 1.000	0.17 [0.08, 0.23]
6	1	1.0000 (s)	—	0.884 / 1.000 / 1.000 / 1.000	below grid
1239	0.3	0.0000 (s); 0.6994 (u); 0.9998 (s)	0.70 [0.64, 0.75]	0.000 / 0.000 / 0.000 / 1.000	0.75 [0.72, 0.77]
1239	0.55	0.0000 (s); 0.7094 (u); 0.9997 (s)	0.71 [0.64, 0.77]	0.000 / 0.000 / 0.166 / 1.000	0.69 [0.63, 0.75]
1239	1	0.0000 (s); 0.7067 (u); 0.9995 (s)	0.71 [0.64, 0.79]	0.000 / 0.000 / 0.422 / 0.944	0.70 [0.61, 0.79]
1569	0.3	1.0000 (s)	—	1.000 / 1.000 / 1.000 / 1.000	—
1569	0.55	1.0000 (s)	—	1.000 / 1.000 / 1.000 / 1.000	—
1569	1	1.0000 (s)	—	1.000 / 1.000 / 1.000 / 1.000	—
91569	0.3	0.0195 (s); 0.3738 (u); 0.9940 (s)	0.37 [0.22, 0.49]	0.178 / 0.656 / 1.000 / 1.000	0.35 [0.28, 0.43]
91569	0.55	0.9958 (s)	—	0.953 / 1.000 / 1.000 / 1.000	—
91569	1	0.9973 (s)	—	1.000 / 1.000 / 1.000 / 1.000	—

Per-cell outcome counts (correct/wrong/undecided of 10 episodes) for the same campaign; intermediate mean final fractions in Fig. 6 of the main text are splits between the two basins:

claim	α	$32x_0 = 4$	$32x_0 = 12$	$32x_0 = 20$	$32x_0 = 28$
6	0.3	2/8/0	10/0/0	10/0/0	10/0/0
6	0.55	3/7/0	10/0/0	10/0/0	10/0/0
6	1	9/1/0	10/0/0	10/0/0	10/0/0
1239	0.3	0/10/0	0/10/0	0/10/0	10/0/0
1239	0.55	0/10/0	0/10/0	2/8/0	10/0/0
1239	1	0/10/0	0/10/0	3/7/0	9/0/1
1569	0.3	10/0/0	10/0/0	10/0/0	10/0/0
1569	0.55	10/0/0	10/0/0	10/0/0	10/0/0
1569	1	10/0/0	10/0/0	10/0/0	10/0/0
91569	0.3	0/9/1	6/4/0	10/0/0	10/0/0
91569	0.55	10/0/0	10/0/0	10/0/0	10/0/0
91569	1	10/0/0	10/0/0	10/0/0	10/0/0

qwen3:8b calibration of the 120 candidate claims found 4 SUPPORTS and 3 REFUTES claims with $|h_q| < 0.4$; a polarity-balanced set of 12 could not be formed, so genuine truth asymmetry remains unidentified in this model as well.

6.2 llama3.3:70b single-agent probe

claim	c_0	β_T	β_F	n	$P(\text{TRUE})$ at $k = 0$
6	1.987	0.636	-0.257	132	0.000
382	3.920	16.536	-0.500	132	0.000
1569	36.547	5.377	-4.482	132	0.000
2070	3.455	1.375	-0.554	132	0.000
90006	0.378	0.487	-0.494	132	0.417
90382	1.739	1.407	-0.748	132	1.000
91569	1.713	1.156	-0.572	132	1.000
92070	3.910	0.203	0.203	132	1.000

Polarity-coded regression across all eight claims (claim fixed effects): $b(\text{TRUE-side message}) = +0.583$, $b(\text{FALSE-side}) = -0.563$, ratio 1.03 with claim-clustered 95% interval $[0.66, 1.64]$ (2,000 resamples of the eight claims; weak ridge 10^{-3} against separation). On the same eight claims and message bank llama3.1:8b gives 1.94 $[1.39, 2.73]$ (over the 17 calibration claims: 1.44 $[1.25, 1.65]$); the 8b–70B difference is 0.90 $[0.00, 1.72]$ and the ratio of ratios 1.87 $[1.00, 3.12]$, so the reduction is at the edge of resolution with eight claims. Per-claim coefficients with $|c_0| > 5$ in the table above reflect separation (accuracy 0 or 1 at $k = 0$) and are reported for completeness. Empty-inbox fields (last column, observed rates over the empty-inbox queries of each claim): the four false originals give $P(\text{TRUE}) = 0.000$ (every answer on the denying and correct side); of the four true rewrites, three give 1.000 and the negated polar-bear rewrite (90006) 0.42, the one near-neutral claim among the eight. Per- k fits (truth-coded, pooled across claims): $\beta_T(4) = 2.41 \rightarrow \beta_T(8) = 1.35$. The positive wrong-side coefficient at $k = 4$ (+0.61) — wrong-side messages slightly *increasing* correct-side probability — is recorded as an observation only.

7 Stage 2c: eight-claim calibration campaign and polarity-origin probe

7.1 Calibration campaign: all 16 cells

Predicted classifications and thresholds come from each claim’s divisive response function fitted to its single-shot queries (Methods); the eight claims were selected from twenty candidates by the pre-registered rule. Prediction intervals are single-agent bootstrap percentiles (200 resamples of

each claim’s queries within design cells) and “frac. bistable” is the fraction of resamples classified bistable. Observed crossings are 50% points of ridge-penalized logistic fits of episode outcomes against x_0 , with episode-bootstrap 95% intervals; $n_c/n_w/n_u$ are correct/wrong/undecided episode counts of 16 per cell.

claim	α	predicted	pred. x_c [95% CI] (frac. bistable)	observed	obs. crossing [95% CI]	$n_c/n_w/n_u$
97	0.3	mono-correct	—	mono-correct	—	16/0/0
97	0.55	mono-correct	—	mono-correct	—	16/0/0
184	0.3	mono-correct	—	mono-correct	—	16/0/0
184	0.55	mono-correct	—	mono-correct	—	16/0/0
590	0.3	bistable	0.559 [0.497, 0.619] (1.00)	bistable	0.560 [0.465, 0.653]	7/9/0
590	0.55	bistable	0.578 [0.516, 0.637] (1.00)	bistable	0.622 [0.536, 0.701]	6/10/0
656	0.3	bistable	0.185 [0.125, 0.267] (1.00)	bistable	0.248 [0.157, 0.341]	12/4/0
656	0.55	bistable	0.119 [0.072, 0.187] (1.00)	bistable	0.120 [0.057, 0.200]	14/2/0
832	0.3	bistable	0.425 [0.364, 0.494] (1.00)	bistable	0.498 [0.413, 0.589]	8/8/0
832	0.55	bistable	0.396 [0.331, 0.466] (1.00)	bistable	0.436 [0.334, 0.531]	9/7/0
1013	0.3	bistable	0.407 [0.344, 0.470] (1.00)	bistable	0.436 [0.335, 0.525]	9/7/0
1013	0.55	bistable	0.409 [0.346, 0.471] (1.00)	bistable	0.373 [0.292, 0.460]	10/6/0
1146	0.3	bistable	0.094 [0.046, 0.157] (0.98)	mono-correct	—	16/0/0
1146	0.55	bistable	0.034 [0.025, 0.101] (0.62)	bistable	0.012 [-0.087, 0.169]	15/1/0
1151	0.3	bistable	0.271 [0.147, 0.395] (0.96)	bistable	0.184 [0.085, 0.285]	13/3/0
1151	0.55	bistable	0.288 [0.178, 0.407] (0.95)	bistable	0.311 [0.208, 0.404]	11/4/1

Table 1: All 16 cells of the eight-claim calibration campaign (qwen3:8b, 256 episodes, zero parse failures). Fifteen cells match in class; the mismatch is claim 1146 at $\alpha = 0.30$, predicted bistable with threshold 0.094 just above the lowest seeded fraction (2/32), observed monostable correct. All eleven predicted thresholds of the cells observed bistable lie inside the observed intervals (mean absolute difference 0.04; Spearman $r_s = 0.96$, claim-permutation $p = 0.004$).

7.2 Polarity-origin probe (D-3)

Judgment-side single-shot measurements on three llama3.1-8b variants with the fixed 8B message bank and a three-shot answer-format prefix applied identically to all variants. The base model does not follow the forced-choice format (parse rate 1.5% in the bare pilot, 4.9% under the prefix), so its row is reported for completeness and carries no statistical weight.

variant	tag (digest)	parsed	$ b_T / b_F $	95% CI
instruct (control)	llama3.1:8b (46e0c10c039e)	5,984/5,984	1.437	[1.250, 1.651]
abliterated	mannix/llama3.1-8b-abliterated (f7d729db9832, Q4_0)	5,984/5,984	1.243	[1.051, 1.471]
base	llama3.1:8b-text-q4_K_M (6f98b5a6e4b7)	294/5,984	(2.10)	not evaluable

Table 2: Per-message assertion asymmetry (claim-fixed-effect logistic regression; claim-clustered bootstrap intervals). The instruct control reproduces the Stage-1 value 1.44 under the added format prefix. Bare affirmation indices S for the four original/negated pairs: instruct 0.92–1.52 (mean 1.22), abliterated 1.13–1.42 (mean 1.24); base cells parsed 2–6 of 64 and are not evaluable.

8 Residual and consistency analyses

8.1 Message statistics by side

claim	side	messages	words/msg	hedges/msg
6	correct	28258	60.7	0.058
6	wrong	35998	59.4	0.419
382	correct	16713	63.5	0.035
382	wrong	47591	62.9	0.116
1569	correct	28115	58.8	0.084
1569	wrong	36093	59.6	0.314
2070	correct	14105	64.5	0.081
2070	wrong	50119	59.8	1.494

8.2 Recorded-message persuasiveness probes

claim	side	message source	$P(\text{correct})$	n
6	correct	bank	1.000	60
6	correct	episode	1.000	60
6	wrong	bank	0.100	60
6	wrong	episode	0.167	60
382	correct	bank	1.000	60
382	correct	episode	1.000	60
382	wrong	bank	0.017	60
382	wrong	episode	0.000	60
1569	correct	bank	1.000	60
1569	correct	episode	1.000	60
1569	wrong	bank	0.100	60
1569	wrong	episode	0.050	60
2070	correct	bank	1.000	60
2070	correct	episode	1.000	60
2070	wrong	bank	0.050	60
2070	wrong	episode	0.167	60

Episode-generated and bank messages are indistinguishable in persuasive force; no message-quality contribution to the high- x_0 residual is detected.

8.3 Repeated exposure and finite time

One-step regression with round interactions gives $\beta_T \times (t-1) = +0.026$ (SE 0.006) and $\beta_F \times (t-1) = -0.004$ (SE 0.003): repeated exposure does not attenuate message influence. Finite-time check on high- x_0 episodes:

α	$32x_0$	n	mean $x_8 - x_7$	frac. interior	frac. still moving
0.2	20	40	-0.008	0.300	0.275
0.2	24	40	-0.009	0.225	0.150
0.3	20	40	-0.012	0.450	0.250
0.3	24	40	+0.001	0.275	0.325
0.38	20	40	+0.001	0.450	0.375
0.38	24	40	-0.028	0.375	0.350
0.45	20	16	-0.020	0.438	0.375
0.45	24	40	-0.008	0.275	0.375
0.55	20	40	-0.009	0.525	0.375
0.55	24	40	-0.002	0.450	0.225
1	20	40	-0.002	0.800	0.575
1	24	40	-0.027	0.725	0.550

8.4 Label residual

Correlation between an episode’s share of TF label assignments and its final state: $r = 0.194$ ($n = 1,008$ episodes) — an order of magnitude below the experimental effects ($\Delta\bar{x}_8 \approx 1$), but nonzero; per-call randomization protects pooled estimates.

8.5 Flip statistics and cascades

Refractory regression: the coefficient of “flipped at $t - 1$ ” on flipping at t is $+0.98$ (claim-cluster SE 0.13) — flips concentrate in particular agents (anti-refractory). Branching ratios:

α	flip events	σ (mean downstream flips)
0.2	9048	3.546
0.3	10129	3.102
0.38	11452	2.771
0.45	10473	2.476
0.55	12301	2.312
1	14625	1.776

Cascade-size distribution in the asynchronous public dataset (GPT-4o-mini, 320 runs; cascades linked by graph adjacency within three events):

cascade size	count
1	9341
2	668
3	130
4	28
5	9
6	1
7	1

91.8% of cascades have size 1; the tail decays geometrically (deeply subcritical).

9 Fidelity check

Ten episodes at $\alpha = 0.30$, $x_0 = 16/32$ re-run with five-sample majority voting: wrong-consensus rate 1.00, undecided 0.00, mean $|x_8 - 0.5| = 0.44$, versus 0.875, 0.125, 0.38 under single-sample judgments — absorption is sharper, not weaker, under the original five-sample protocol.